PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting

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pup3dgs.github.io

Motivation

How can we compress pretrained 3D Gaussian Splatting (3D-GS) models by 10× while preserving image quality?

We introduce a mathematically principled per-Gaussian **pruning score** and an effective **pruning pipeline** that together yield surprisingly strong results.

Method

Pruning Score

We compute a per-Gaussian pruning score U_i as the log determinant of the Hessian of the L_2 reconstruction error for Gaussian $\mathcal{G}_i$, where $\mathcal{P}_{gt}$ is the set of all training poses and $I_{\mathcal{G}}(\phi)$ is the rendered view for pose ϕ:

$$U_i \approx \log\left|\nabla^2_{\mathcal{G}_i} L_2\right| \approx \log\left|\sum_{\phi\in\mathcal{P}_{gt}} \nabla_{\mathcal{G}_i} I_{\mathcal{G}}(\phi) \nabla_{\mathcal{G}_i} I_{\mathcal{G}}(\phi)^T\right|.$$

We find that the Gaussian mean μ_i and scaling s_i parameters produce an effective **spatial sensitivity pruning score**:

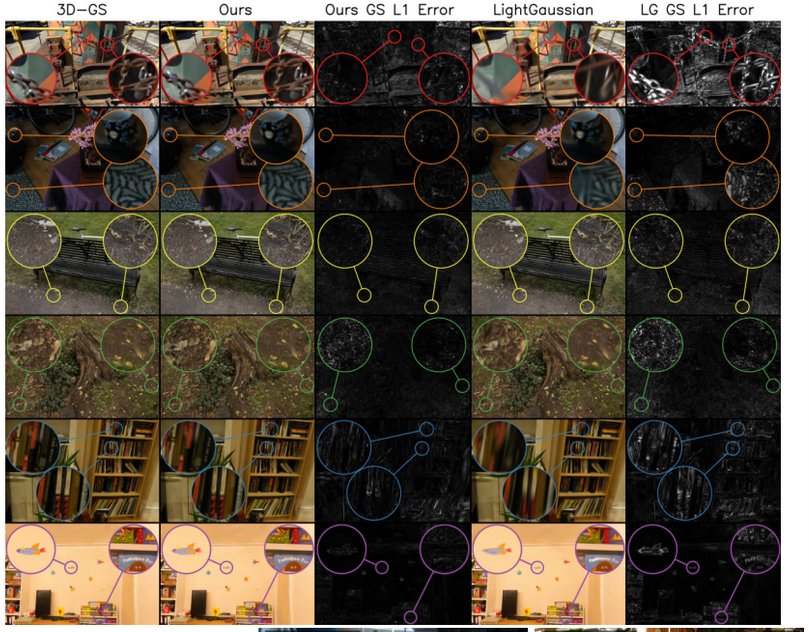
$$U_i = \log\left|\sum_{\phi\in\mathcal{P}_{gt}} \nabla_{\mu_i,s_i} I_{\mathcal{G}}(\phi) \nabla_{\mu_i,s_i} I_{\mathcal{G}}(\phi)^T\right|.$$

Pruning Pipeline

(1) We **prune 80% of Gaussians** and fine-tune for 5,000 iterations, then
(2) **prune 50% of Gaussians** and fine-tune for 5,000 more iterations.

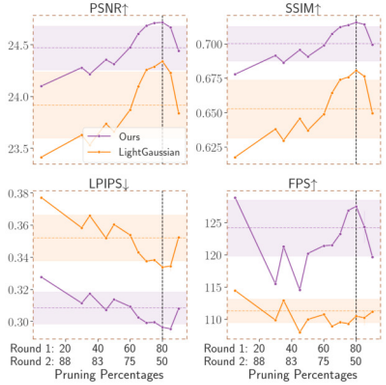
In total, we **prune 90% of Gaussians from the pretrained model**.

Comparison



PUP 3D-GS **retains more fine details** than previous methods when **pruning 90% of Gaussians** from 3D-GS.

Results



When applied to pretrained 3D-GS models,

PUP 3D-GS achieves **3.56× FPS** and **10× smaller model sizes**

while preserving image quality and salient foreground information.

Datasets	Methods	PSNR ↑	SSIM ↑	LPIPS ↓	FPS ↑	Size (MB) ↓
MipNeRF-360	3D-GS	*27.47*	*0.8123*	*0.2216*	*64.07*	*746.46*
	LightGaussian	26.28	0.7622	0.3054	162.12	74.65
	Ours	26.67	0.7862	0.2719	204.81	74.65
Tanks & Temples	3D-GS	*23.77*	*0.8458*	*0.1777*	*97.86*	*433.24*
	LightGaussian	23.08	0.7950	0.2634	329.03	43.33
	Ours	22.72	0.8013	0.2441	391.10	43.33
Deep Blending	3D-GS	*28.98*	*0.8816*	*0.2859*	*66.79*	*699.19*
	LightGaussian	28.51	0.8675	0.3292	234.10	69.92
	Ours	28.85	0.8810	0.3015	301.43	69.92

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